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Perceiving absent information from contemporary art to enhance awareness

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Abstract

We performed a preliminary study to explore the mechanisms underlying the perception and thinking related to contemporary art. In this experiment, participants were asked to describe two contemporary artworks: one was displayed in a gallery and the other was an abstract painting. The perceptual processing of absent information was discussed in terms of predicate logic and the landscape montage technique.

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1. Introduction

What do people perceive and how does contemporary art stimulate their thinking?

Kanda [1] proposed that the human intellectual production process comprises three stages (Fig. 1), and perception is positioned at the forefront of these stages. This study focused on the transition from stage 1 to stage 2. The analysis and logical thinking for problem-solving used for the management consultancy work undertaken by the authors predominantly align with stage 2. However, the authors have unconsciously utilized creative perspectives and unique interpretations based on perception, thus operating prior to stage 2. The key distinction between perception and thinking lies in their controllability; perception represents a stage that cannot be controlled by the individual.

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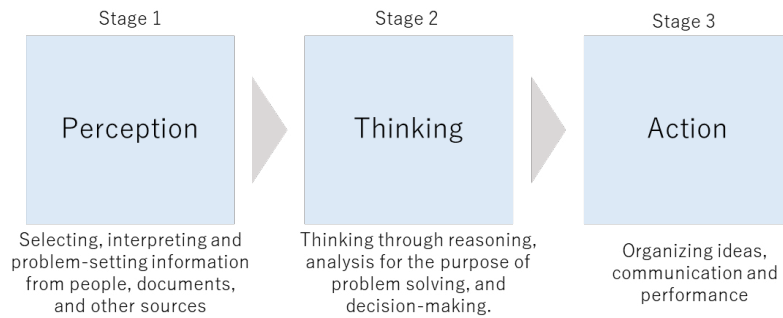


Fig. 1. Intellectual production process [1].

Perception may be the most important intellectual production process, strongly influencing subsequent stages. However, because perception is beyond conscious control, individuals often subconsciously perceive and merge their past experiences and knowledge when perceiving external stimuli, such as contemporary art. The act of painting a rural landscape on a blank canvas may also trigger subconscious recollection of past memories. The authors propose that individuals perceive aspects beyond physical visibility, namely absent information. In this context, the present pilot study investigated the relationship between contemporary art and an individual's perception.

2. Approaches to understanding art

Adelson [2] proposed two mechanisms for understanding art: bottom-up processing and top-down processing. When visual information, such as artwork, is input, it undergoes two types of processing. First, in bottom-up processing, the image of the artwork is interpreted by eliciting main components of the physical world, such as contours, line intersections, and joints. This process leads to the recognition of objects, human faces, spatial positioning, and (ultimately) their beauty, mirroring how infants recognise human faces. In abstract paintings, bottom-up processing involves identifying features of basic shapes and colours that are in a state of separation.

Second, top-down processing involves making inferences about meaning from visual information. The brain attempts to test these hypotheses by formulating guesses. At this stage, psychological contextual dependence on individual memories and experiences leads to different understandings of meaning of the same object for different individuals [3]. Therefore, top-down processing is a creative process of linking components together while shifting the focus of attention to the components of visual information.

In many abstract paintings, colour, line, shape, and brightness are separated. Therefore, a neuroscientific approach has been discussed by integrating bottom-up perceptual processing and top-down processing to contextually reconstruct the components [4].

Past learned memories and experiences undergo transformation into representations of objects through bottom-up processing (visual processing at lower and intermediate levels), particularly in the inferior temporal cortex of the human brain. In the inferior temporal cortex, multiple memory objects are integrated, stored, and anchored in memory structures within the medial temporal lobe. Numerous experiments involving brain activation have demonstrated that the visual pathways consist of interactive interdependencies among the prefrontal cortex (working memory), medial temporal lobe, and inferior temporal cortex. However, the perception of images retrieved from memory is mediated by top-down processing originating from the prefrontal cortex [5].

Tadaki and Abe [6] demonstrated that concrete painting-like appreciation occurs through the appreciation of abstract paintings. The previous study utilized a statistical approach, whereas the present study focuses on structuring predicate logic and comparing it with the landscape montage technique, thereby exploring the mechanism of perception and its generation from contemporary art.

3. The landscape Montage Technique

The landscape montage technique is a form of painting therapy (art therapy), originally developed by Hisao Nakai in 1969 and reported in 1970, which has the practical aim of exploring the possibilities and applicability of treatment for schizophrenic patients through drawing [7]. In this procedure, the therapist frames the four perimeters on a canvas paper using a signet pen in front of the patient, then hands the paper and pen to the patient. Next, the therapist instructs the patient not to evaluate the drawing quality, emphasising that it is their own landscape and that any landscape can be created. The patient then proceeds to draw in the following order.

- Midium view group: houses → trees → people
- Others: flowers → animals → stones and rocks → things you think are missing or want to add

Once an image is drawn on the canvas, it cannot be moved, and the authors suggest the “distortion” resulting from the restriction of freedom might stimulate thought [8].

In this regard, the authors believe that the ability to perceive absent information (i.e., invisible objects) is a common skill required to appreciate contemporary art. As the name implies, the landscape composition method evokes landscapes such as the earth and the horizon in the unseen realm, whereas contemporary art also explores the unseen but not necessarily tangible objects.

4. Experiment

As an external input, two works of contemporary art were selected: a minimalistic[†] spatial expression by the American sculptor Fred Sandback and an abstract painting by the Dutch abstract painter Mondrian[‡]. A simple experiment was conducted in which participants were presented with these artworks and asked to freely describe their thoughts and feelings, including through drawing. The descriptions were used to explore the process of shifting from perception to thought[§]. The experiment included 40 second-year university students in Japan; it was conducted between May 15 and May 23, 2023. No time limit was set for the working period to avoid hindering creative thinking.

This paper has focused on contemporary art because it is generally more challenging to interpret the meanings of shapes and colours compared with realistic art; thus, it is more conducive to explorations of the mechanisms of creative perception. Due to individual differences in experience and knowledge, various cases can be envisioned from perception to thought. The authors believe that if a common intellectual production process [1] can be observed through the experiment, the appreciation of contemporary art will contribute to a deeper understanding of the creative process.

The analysis method aims to extract absent information by simplifying sentences^{**} written by individuals and expressing them in terms of formal logic, particularly predicate logic. In this context, absent information does not pertain to one’s impressions of a particular work but rather to what is considered contextually significant as newly generated opinions or statements. The discussion will also integrate the concept of the landscape montage technique.

5. Results

Analysis of data from the 40 respondents revealed considerable variation in the number of letters, ranging from 152 to 1,305. Most of the content was based on past memories and experiences, consistent with the top-down approach.

[†] A style of expression that minimises decorative flourishes, rather than elaborating them in pursuit of perfection (Wikipedia).

[‡] In this paper, Fred Sandback’s work is described as contemporary art. Mondrian’s work is also described as contemporary art but is particularly related to abstract painting.

[§] The act of individuals describing their responses during the experiment is referred to as stage 3 in Figure 1.

^{**} The output text is presented in [10].

However, several respondents demonstrated a tendency to combine, transform, or rotate elements to form new contexts, indicating a bottom-up approach.

All participants exhibited certain leaps in thinking. A representative case, expressed in formal logic, is presented below, beginning with the description of Fred Sandback's design space.



Fig. 2. Space by Fred Sandback [11].

The following are excerpts from the generated text.

Participant F

$\exists xFx$: there exists an x (triangle) satisfying “ x (triangle) overlaps”

$G(x, y)$: x can be interpreted as y (cultural overlapping)

$R(y, z)$: y (overlapping cultures) generates z (idea)

$\exists xFx \wedge G(x, y) \rightarrow R(y, z)$

Absent information: overlapping layers of culture

Participant M

$\exists xFx$: there exists an x (triangle) satisfying “ x (triangle) is aligned at regular intervals”

$G(x, y)$: x can be interpreted as y (space)

$G(x, z)$: x can be interpreted as z (no depth)

$G(x, a)$: x can be interpreted as a (trick)

$R(i)$: “ i is mistaken” holds for i (information obtained at the moment)

$\exists xFx \wedge G(x, y) \wedge G(y, z) \wedge G(x, a) \rightarrow R(i)$

Absent information: risk of being mistaken due to trickery

As described above, many patterns of thinking were identified, such as A and B is C . Accordingly, there is an understanding of the form, followed by its interpretation and expression of an opinion. In a later interview, participant F said that she recalled taking a class on cross-cultural management theory.

Conversely, some expressions were not based on newly formed opinions, such as “I had an impression” or “looks like.” These expressions do not provide new meanings but only describe the perceptions about features such as the shape. In this case, no contextual information is absent. For example:

$\exists xFx$: there exists an x (triangle) satisfying “there is more than one x (triangle)”

$G(x, y)$: x appears to be y (smaller toward the back)

$G(x, y, z)$: y (smaller with increasing depth), x (plural triangles), appears to be z (sucked in)

$\exists xFx \wedge G(x, y, z)$

Next, we discuss the abstract paintings by Mondrian.

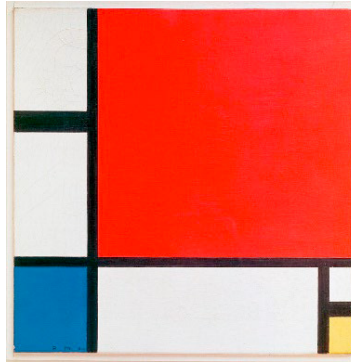


Fig. 3. Red, Blue, and Yellow Composition by Mondrian [12].

Participant F

[P (x1) = Q (y1)]: assume x1 (red) has a similar meaning to y1 (bully)

[P (x2) = Q (y2)]: x2 (blue) is assumed to have a similar meaning to y2 (bullied person)

[P (x3) = Q (y3)]: assume x3 (yellow) has a similar meaning to y3 (teacher not involved in student's problems)

[P (x4) = Q (y4)]: x4 (white) is assumed to have a similar meaning to y4 (bystander)

[P (x5) = Q (y5)]: x5 (black) is assumed to have a similar meaning to y5 (cause)

Absent information: social issue of bullying

As mentioned above, colour is semantically easy to interpret. In the post-experiment interviews, participants stated that the idea of bullying was due to the recall of recent news.

Participant M

[P(x1)=Q(y1)]: x1 (red) is assumed to mean the same as y1 (weak majority opinion)

[P(x2)=Q(y2)]: x2 (blue) is assumed to mean the same as y2 (minority opinion)

[P(x3)=Q(y3)]: x3 (yellow) is assumed to mean the same as y3 (wait and see)

[P(x4)=Q(y4)]: x4 (white) is assumed to mean the same as y4 (no opinion)

G(z): G(z): we should respect z (the other person's opinion).

[P(x) ∧ Q(y)] → G(z)

Absent information: addressing disagreements

However, in some cases, such as those discussed next, the statements are merely based on the perceptions of participants, such as how they appear, rather than on memory or knowledge. Therefore, no contextual information is missing.

Participant T

∃ xFx: there exists x (square) satisfying “x (square) is divided by a black line”

G (x, y): x appears to be y (dyed in colour)

G (x, y, z): y (dyed in colour) and x (square), which is partitioned by black lines, appears to be z (feeling of openness)

∃ xFx ∧ G (x, y, z).

We found that the basic form is [P(x) ∧ Q(y)] → G(z), although certain expressions are more complex, with ternary and quaternary predicates rather than binary predicates.

6. Discussion

This chapter discussed absent information, followed by our experimental results with reference to the landscape montage technique, and (finally) perceptual processing.

First, the absent information is discussed. The logical structure of the response content was comparatively simpler in the case of the artworks by Mondrian, involving the “meaning of colour $\Rightarrow$ expression of opinion,” where multiple colours were combined. This simpler structure may arise from the fact that, among abstract paintings, expressions with plentiful colours are easier to interpret in terms of meaning compared with expressions involving only shapes. For example, red evokes images of real objects (e.g., apples) while tending to evoke absent information such as passionate feelings, even if those feelings do not tangibly exist. Conversely, in the case of Fred Sandback’s artwork, shapes such as triangles may evoke images of mountains, but the generation of context solely from the shape itself is comparatively more challenging because it requires the addition of absent information in the viewer’s mind. Moreover, without a contextual reference, the simple shape itself is difficult to describe using adjectives such as beautiful or heavy.

Second, a discussion with reference to the landscape montage technique is presented. In this technique, “watching” is examined through syntagmatic choice, and “seeing” is analysed through paradigmatic choice [13]. Syntagmatic choice refers to the selection made during the formation of a sentence, where the verbs complement each other to create a coherent whole. For example, a specific verb is selected by a particular participant. Paradigmatic choice involves selecting one of several synonymous adjectives to describe a noun.

Takashima [14] stated the following.

The process of syntagmatic choice would, in a sense, be the same as the process of creating the framework of a story. (omitted) Only through 'making what one experiences into a story that makes sense to oneself and finding a path in it' is [one] able to put it into his or her own mind. (omitted) The connections between the items may show how the artist perceives the connections between the people and things that inhabit the world, and what path he sees in the way the world is and in his own experience.

Most patterns observed were aimed at forming an opinion linked to memory and knowledge or at conveying an impression of the shape. This disparity could represent the difference between what is often regarded as “watching” and what is often regarded as “seeing”; it reflects a difference in the ability to discern what cannot be seen (i.e., information that is absent).

When transitioning from perception to contemplation of contemporary art, the process of filling in the gaps (connecting elements such as shapes and colours) is influenced by each individual’s background, knowledge, and experiences, thereby producing a distinct narrative. Both syntagmatic and paradigmatic choices are believed to be intertwined in the case of both artworks in the experiment.

Third, the discussion extends to perceptual processing. Contemporary art often presents formal elements such as colour, line, shape, and brightness, which are isolated from one another. Consequently, viewers must engage in both bottom-up perceptual processing and top-down processing to contextualise these formal elements. The experimental findings align with prior studies, highlighting the roles of memory and experience in top-down processing. Indeed, some respondents (students) sporadically referenced their experiences at the university, further supporting this notion.

Additionally, thoughts were developed on the concepts of “rotation”, “combination (of colours),” or “transformation (of forms)” of art forms. For example, as shown in the figure below.

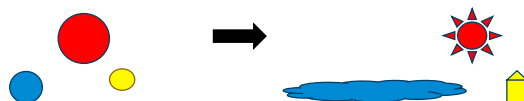


Fig. 4. Transformation of shapes by the participant.

